Roll No. ....

TBT-801

B. TECH. (BIOTECHNOLOGY) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024

OMICS TECHNOLOGIES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How next generation sequencing platforms are different from conventional platforms ? (CO1)

   (b) Write short notes on any *two* of the following : (CO1)

   (i) DNA foot printing

   (ii) Exon-Intron trapping

   (iii) S-1 mapping

   (c) How clone contig approach of genome sequencing is different from shot gun ? (CO1)

P. T. O.

2. (a) Give a comparative account of forward and reverse genetics approaches of the functional genomics. (CO2)

   (b) How random mutations can be tagged in forward genetic approach ? (CO2)

   (c) Discuss the methodology and applications of RNA interference in gene silencing. (CO2)

3. (a) Discuss characteristic properties of Yeast as a preferred model organism in laboratories. (CO3)

   (b) Write short notes on any *two* of the following : (CO3)

      (i) Metagenomics

      (ii) Toxicogenomics

      (iii) Pharmacogenomics

   (c) What is knockout mouse ? Discuss its importance in drug research. (CO3)

4. (a) Write down the principle and applications of mass spectroscopy in proteomics. (CO4)

   (b) What is proteomics ? Describe role of 2D SDS-PAGE in proteomics. (CO4)

   (c) How protein-protein interaction is studied in vivo ? (CO4)

5. (a) What is transcriptomics ? How transcriptomics can be studied ? (CO5)

   (b) What is quantitative RT-PCP Write its application in the diagnosis. (CO5)

   (c) Discuss microarray as a system to analyze transcriptome ? Discuss its advantages and limitations. (CO5)

Roll No. ....

TBT-802(a)

B. TECH. (BIOTECHNOLOGY) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

PROJECT MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe Market and Demand Analysis with the help of an example. (CO2)

(b) Describe Environmental Appraisal and the factors that influence it. (CO2)

(c) What is Project Management ? What are the roles and responsibilities of a Project Manager ? (CO2)

2. (a) What is WBS ? How WBS helps the Project Manager ? Also describe the hierarchical breakdown of WBS. (CO2 & CO3)

(b) Define Project Stakeholders ? Who are the key important stakeholders in a Project. (CO2 & CO3)

(c) Write a note on Risk Management in Project. (CO2 & CO3)

3. (a) Describe Project cost. What are the different types of project cost ? Describe each of them ? (CO3)

(b) Describe Project Network. Create Project Management network diagram. Also mention the benefits of network Diagram. (CO3)

(c) Write short notes on CPM and PERT. (CO3)

4. (a) Define Project Closure and why it is important ? (CO1 & CO4)

(b) What do you understand by managing a team in project management ? What are the methods of managing a project team ? (CO1 & CO4)

(c) Define virtual Project Management. What are the benefits of virtual project management ? (CO1 & CO4)

5. (a) Define the term project team pitfalls. How can a Project Manager avoid them ? (CO5)

(b) Write short notes on Project Boundaries and Scope of the Project. (CO5)

(c) Write short notes on SBCA (Social Cost Benefit Analysis) in a Project. (CO5)